A free endpoint for odd orthogonal strings in dimension five

Human Compression project: F001 working component

This is a construction and exceptional-field component, not yet a complete recovery of Brooksbank's rank spectrum theorem. We prove existence in odd fields of sufficiently large order without the primitive-parameter density assertion in the supplied preprint. Exact finite certificates supply the remaining odd construction cases and the exceptional field of order three. The upper bound on ranks for the other odd fields is a separate obligation. This working edition has not received distinct review.

1 Classical inputs and the three-dimensional step

Let $F = \mathbf{F}_q$ have odd order, let Q be a quadratic form, and write $B(u, v) = Q(u+v)-Q(u)-Q(v)$. For $Q(u) \neq 0$ its reflection is

$$s_u(v) = v - \frac{B(v, u)}{Q(u)}u.$$

For a subspace U write $K_\square(U)$ for the group generated by the reflections whose roots lie in U and have nonzero square norm; all such groups act on the same ambient space. We retain the classical spinor-norm generation theorem: on a nondegenerate space, these reflections generate the kernel of the spinor norm. We also retain the three-dimensional identifications $\mathrm{SO}_3(q) \cong \mathrm{PGL}_2(q)$ and $\Omega_3(q) \cong \mathrm{PSL}_2(q)$, and Dickson's subgroup classification in the following exact form, as stated in Leemans–Moerenhout–O'Reilly-Regueiro, *Projective linear groups as automorphism groups of chiral polytopes*, supplied twenty-five-page edition, Proposition 2.3, p.5. A subgroup of $\mathrm{PSL}_2(q)$ or $\mathrm{PGL}_2(q)$ is either contained in a point stabilizer, a torus normalizer, a subfield subgroup, one of the exceptional groups A_4, S_4, A_5, or contains $\mathrm{PSL}_2(q)$. The full proofs of these classical inputs remain external acquisition costs. The string end-parabolic criterion is the already acquired McMullen–Schulte criterion.

Put $\epsilon = \chi(-1)$ and $N = (q + \epsilon)/2$, where χ is the quadratic character. Thus N is odd. Call a nondegenerate plane L suitable if $K_\square(L)$ is dihedral of order $2N$. Its rotation subgroup then has order N.

Lemma 1. *Suppose $q \geq 13$. Let $T = L + \langle x \rangle$ be a three-space containing a suitable plane L, where $Q(x)$ is a nonzero square and $x \notin L \cup L^\perp$. Then*

$$\langle K_\square(L), s_x \rangle = K_\square(T).$$

The conclusion holds whether or not T is nondegenerate, provided the ambient space is nondegenerate.

Proof. First suppose T nondegenerate. Let $D = K_\square(L)$ and let $C \leq D$ be its cyclic subgroup of order N. Under $g \mapsto \det(g)g$, the group $K_\square(T)$ maps onto either $\mathrm{PSL}_2(q)$ or $\mathrm{PGL}_2(q)$, with kernel $\{1, -1_T\}$ or 1, respectively. This follows directly from the determinant and spinor norm: the first case occurs precisely when the spinor norm of -1_T is square. The map is injective on D, since D fixes the nonzero perpendicular line $L^\perp$ pointwise. Hence its image contains a dihedral group of order $2N$.

Here $N > 5$ and $N > \sqrt{q} + 1$. The exceptional groups and proper subfield groups cannot contain an element of order N: the semisimple orders in $\mathrm{PGL}_2(q_0)$ are at most $q_0 + 1$, where

$q_0 \le \sqrt{q}$. A point stabilizer cannot contain this dihedral group either. Its image on its diagonal quotient is abelian and injective on C, whereas a reflection of D inverts C. Dickson's classification leaves only the torus normalizer and a group containing $\mathrm{PSL}_2(q)$.

The normalizer of C in the three-dimensional orthogonal group preserves L and $L^\perp$: these are its nontrivial and trivial constituents. A reflection preserves L precisely when its root lies in $L \cup L^\perp$. Thus s_x excludes the normalizer. If the target is $\mathrm{PGL}_2(q)$, the image of s_x has nonsquare spinor norm, so the image is the whole target rather than its index-two subgroup. If the target is $\mathrm{PSL}_2(q)$, the full group upstairs is $\langle -1_T \rangle \times \mathrm{PSL}_2(q)$; a subgroup mapping bijectively onto the second factor is a graph of a homomorphism to C_2. Perfection of $\mathrm{PSL}_2(q)$ makes that graph have determinant one, contrary to the presence of s_x. In both cases the generated group is $K_\square(T)$.

Now suppose T degenerate. Write $T = L \perp \langle z \rangle$ with $Q(z) = 0$, and choose a singular partner w with $B(z, w) = 1$, perpendicular to L. The group supported on T has translation kernel $\{t_u : u \in L\}$, where

$$t_u(y) = y - B(y, u)z \ (y \in L), \quad t_u(z) = z, \quad t_u(w) = w + u - Q(u)z.$$

These maps multiply by $t_u t_v = t_{u+v}$. The subgroup generated by square-norm reflections is $L^+ \rtimes D$: the displayed translations have odd order and trivial spinor norm, and products of the reflections in u and $u + az$ provide all scalar multiples of every square-norm u as translation parameters. Such roots span L; alternatively this follows from the irreducibility below. Writing $x = u + az$, with $a \neq 0$, shows that $s_x s_u$ is a nonzero translation. Its D-conjugates supply the additive span of a nonzero D-orbit.

That span is all of L, even over the prime field. Here are the field details. Write $q = p^e$, and let λ be an eigenvalue of a generator of C, of order N. In the nonsplit case its degree over $\mathbf{F}_p$ is $2e$: any proper degree divides $2e$ and is at most e; degree e would put N into $q-1$, and a smaller degree is impossible because $N > p^{e-1} - 1$. In the split case λ has degree e. The two eigenvalues are not conjugate over $\mathbf{F}_p$, since such conjugacy would give $N \le p^{e-1} + 1 \le q/3 + 1 < N$ for $q > 9$. Their two irreducible prime-field constituents are exchanged by the reflections in D. Thus D acts irreducibly on L over $\mathbf{F}_p$ in either case. The nonzero translation therefore supplies the whole kernel and proves the assertion. □

Lemma 2 (Saturation from a suitable plane)**.** *Suppose $q \ge 13$, V is nondegenerate of dimension at least four, and W is a nondegenerate hyperplane. Let L be a suitable plane not contained in W, whose intersection with W has square norm. Then $\langle K_\square(W), K_\square(L) \rangle = K_\square(V)$.*

Proof. It suffices to obtain s_y for every square-norm root $y \notin W \cup L$. Put $T = L + \langle y \rangle$ and $M = T \cap W$. The two-space M contains the square-norm line $L \cap W$. If M is nondegenerate, it has $(q-1)/2$ or $(q+1)/2$ square-norm projective points. If it is degenerate, the presence of that line shows it has q such points. There are therefore at least six. At most one lies in L, and at most one in $L^\perp \cap T$. Choose another, represented by x. Its reflection is already in $K_\square(W)$; Lemma 1 gives $K_\square(T)$ and in particular s_y. □

2 A path with one free bond

Consider the five-space with quadratic form

$$Q(X) = \sum_{i=1}^{5} X_i^2 + \eta X_1 X_2 + t(X_2 X_3 + X_3 X_4 + X_4 X_5), \tag{1}$$

where $t\eta \neq 0$. Suppose every consecutive coordinate subspace is nondegenerate, and each of the final three adjacent products of coordinate reflections has order $N = (q + \chi(-1))/2$. The corresponding two-space groups are then precisely their $K_\square$ groups: two reflections of square norm generate a dihedral group of order $2N$, the full spinor kernel in that plane.

Proposition 3. *For $q \geq 13$, the five coordinate reflections of (1) form a string C-group and generate $K_{\square}(V)$. If the determinant of the symmetric matrix representing Q is a nonzero square, their negatives form a rank-five string C-group for $\Omega_5(q)$. It has rank-four and rank-three contractions.*

Proof. Every consecutive triple has a suitable pair. Its other root is outside that plane and is not perpendicular to it, so Lemma 1 gives the full $K_{\square}$ group on the triple. For an interval $v_1, \ldots, v_b$ of length $b \geq 4$, its suffix $W = \langle v_2, \ldots, v_b \rangle$ has full $K_{\square}(W)$ by induction. The plane $L_0 = \langle v_2, v_3 \rangle$ is suitable. Conjugate it by s_{v_1} to obtain a suitable plane L not contained in W. The line $L \cap W = \langle v_3 \rangle$ has norm one. Lemma 2 therefore gives the full group on the interval. This proves generation for all intervals of length at least three.

Every two-generator subgroup is a dihedral C-group. For the intersection step, an isometry supported on a nondegenerate subspace U fixes $U^{\perp}$ pointwise; conversely, its restriction to U determines it. Thus for nondegenerate $U, W, U \cap W$,

$$K_{\square}(U) \cap K_{\square}(W) = K_{\square}(U \cap W).$$

The same assertion on a square-norm one-space is its order-two reflection group. For a triple this bounds the intersection of the two dihedral end groups by its middle reflection. For length four the common middle plane is suitable, and for length five the common middle three-space has already been generated. Induction with the end-parabolic criterion proves the full intersection property. Nonadjacent coordinate roots are perpendicular, giving the string relations.

The spinor norm of -1_V is the determinant of Q modulo squares. Under the determinant hypothesis it lies in $K_{\square}(V)$, and $g \mapsto \det(g)g$ maps this group onto $\Omega_5(q)$ with kernel $\langle -1_V \rangle$. Each end parabolic fixes a nonzero perpendicular line pointwise, so the map is injective there. We must also exclude a new intersection after quotienting. Let a and b span the perpendicular lines to the two end hyperplanes. Their pairing is nonzero: the corner entry of the inverse tridiagonal form matrix is its four nonzero adjacent entries multiplied together, divided by its nonzero determinant. If elements h_0, h_4 of the two end groups had $h_0 = -h_4$, then h_0 would fix a and negate b. This contradicts $B(a, b) \neq 0$ in odd characteristic. Thus the intersection of the two image groups is exactly the image of their old intersection. Their individual markings are faithful images of C-groups, so the end-parabolic criterion proves the assertion. Their adjacent product orders equal those of the original reflections. The first adjacent pair is noncommuting because its independent roots have nonzero pairing η; the remaining pairs have order $N > 2$. The marking is therefore irreducible. The third and fourth bonds have odd order N, so two applications of the acquired rank-contraction theorem give ranks four and three. □

3 Uniform choice of parameters

Theorem 4. *If $q \geq 33$ is an odd prime power, then $\Omega_5(q) \cong \mathrm{PSp}_4(q)$ has string C-group representations of ranks five, four and three.*

Proof. Let $\epsilon = \chi(-1)$, $N = (q + \epsilon)/2$, and take an element μ of exact order N in $\mathbf{F}_q^{\times}$ when $\epsilon = -1$, or in the norm-one torus of $\mathbf{F}_{q^2}^{\times}$ when $\epsilon = 1$. Put $t = \mu + \mu^{-1} \in F$ and $x = t^2/4$. The product of reflections on a plane with norm-one roots and pairing t has characteristic polynomial $X^2 - (t^2 - 2)X + 1$ and eigenvalues μ^2, μ^{-2}. It therefore has order N. Also

$$1 - x = -\frac{(\mu - \mu^{-1})^2}{4}$$

is a nonsquare in F in both torus cases. In particular t and $1 - x$ are nonzero. The equality $1 - 2x = 0$ would imply $\mu^4 = -1$, impossible for the odd-order element μ.

Choose μ so that also $A = 1 - 3x + x^2 \neq 0$. The equation $A = 0$, multiplied by $16\mu^4$, is

$$\mu^8 - 8\mu^6 - 2\mu^4 - 8\mu^2 + 1 = 0. \tag{2}$$

There are at most eight roots, while $\varphi(N) > 8$ since $N \geq 17$ is odd. For completeness, the odd integers with Euler totient at most eight are exactly $1, 3, 5, 7, 9, 15$: their prime divisors can only be $3, 5, 7$, and multiplicativity and the prime-power formula give this list. Hence a permissible primitive μ exists without any density estimate.

Now let η vary in F. The determinants of the three proper prefixes of length two, three and four are, with $y = \eta^2/4$,

$$1 - y, \qquad 1 - x - y, \qquad 1 - 2x - y(1 - x). \tag{3}$$

The suffix determinants of lengths two, three and four are $1 - x$, $1 - 2x$, and A, all already nonzero. The full determinant is

$$\Delta = A - B\eta^2, \qquad B = (1 - 2x)/4 \neq 0. \tag{4}$$

For $A, B \neq 0$, the elementary character identity

$$\sum_{\eta \in F} \chi(A - B\eta^2) = -\chi(-B)$$

follows by counting $(u - v)(u + v) = c$ for $c \neq 0$. At most two summands are zero. Consequently at least $(q - 3)/2$ values of η give a nonzero square Δ. Discard $\eta = 0$ and the at most six roots in (3). Since $(q - 3)/2 > 7$, a value remains. Every consecutive subspace is then nondegenerate, and Proposition 3 applies. □

4 Finite parameters and the four small fields

The hypothesis $q \geq 33$ in Theorem 4 is used only to choose the parameters. Proposition 3 applies already for $q \geq 13$. The following table supplies the intervening fields. Here z_{25} is a root of $X^2 - X + 2$ over $\mathbf{F}_5$, and z_{27} is a root of $X^3 - X + 1$ over $\mathbf{F}_3$, chosen as in the accompanying GAP field model. Equivalently these are its Conway generators $Z(25)$ and $Z(27)$; the certificate states the exact powers.

q	13	17	19	23	25	27	29	31
t	10	14	3	14	z_{25}^{11}	z_{27}^{16}	14	16
η	1	3	4	1	z_{25}^{2}	z_{27}^{4}	4	19

(5)

The finite checker verifies all fifteen interval determinants, the square class of the full determinant, and the exact order $(q + \chi(-1))/2$ of the last three products for every row. Thus no group enumeration is needed for these eight fields.

For $q = 5, 7, 9, 11$ use

$$Q(X) = \sum_{i=1}^{5} X_i^2 + \sum_{i=1}^{4} c_i X_i X_{i+1}$$

with the following coefficients. The element $z \in \mathbf{F}_9$ satisfies $z^2 = z + 1$ and has order eight.

q	(c_1, c_2, c_3, c_4)	adjacent orders	$\lvert\Omega_5(q)\rvert$
5	$(1, 1, 2, 1)$	$(3, 3, 5, 3)$	4 680 000
7	$(1, 1, 1, 2)$	$(3, 3, 3, 7)$	138 297 600
9	$(z, 1, 1, z)$	$(5, 3, 3, 5)$	1 721 606 400
11	$(1, 1, 1, 3)$	$(3, 3, 3, 5)$	12 860 654 400

(6)

In these four cases some proper intervals may be degenerate, so we use the complete finite certificate in place of Proposition 3. Let M be the symmetric matrix of Q. The reflection in the ith coordinate root is the row-action matrix $R_i = I - 2(Me_i)e_i^{\mathsf{T}}$; its negative has determinant

one. The certificate verifies $R_i M R_i^{\mathsf{T}} = M$ and that $\det M$ is a nonzero square. Hence the negative reflections lie in $\Omega_5(q)$ by their spinor norms.

Their permutation action is computed on every isotropic projective point, normalized by making the first nonzero coordinate one. There are $(q+1)(q^2+1)$ such points, namely 156, 400, 820 and 1464 in the four cases. This action is faithful on the group of negative reflections. The certificate exhibits five independent isotropic representatives and verifies that the supports of the remaining representatives connect all five coordinates in that basis. A linear map fixing all their projective points is diagonal in the chosen basis, and those connected supports make all five diagonal entries equal. A scalar isometry of determinant one in odd dimension is the identity. This proves the required faithfulness without a faithfulness assumption about a permutation library model.

The permutation groups have the orders in (6), equal to $q^4(q^2-1)(q^4-1)/2$, the classical order of $\Omega_5(q)$. They are therefore the full groups. Each tuple has five involutions, the displayed adjacent orders, and the string commutations. For all $32^2 = 1024$ ordered subset pairs the certificate verifies the intersection condition. Containment pairs are immediate; other pairs are checked by exact group intersection, with the symmetric pair reused. The last two bonds in every row have odd order, so the acquired contraction theorem gives ranks four and three as well.

Theorem 5. *For every odd prime power $q \geq 5$, the group $\mathrm{PSp}_4(q) \cong \Omega_5(q)$ has string C-group representations of ranks three, four and five.*

Proof. Combine Theorem 4, the checked rows (5), and the four exact constructions (6). □

5 The exceptional group of order 25920

We give a separate exhaustive certificate that $\mathrm{PSp}_4(3)$ has no string C-group representation. To identify the group without naming a library group, use the alternating form

$$J = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

on $\mathbf{F}_3^4$ and the transvections $w \mapsto w + (wJv^{\mathsf{T}})v$ for the forty normalized projective vectors v. Their action on these same forty points has order 25920, half the classical order of $\mathrm{Sp}_4(3)$. Thus it is the full projective symplectic group. It is simple, as also verified by the finite checker, and has 315 involutions in two classes of sizes 45 and 270.

Here is the complete extension procedure. A node is an ordered prefix $(s_1, \ldots, s_r)$ that is already a connected string C-marking of its generated subgroup H. Fix one representative for s_1 in each involution class. At a node take the involutions

$$x \in C_G(s_1, \ldots, s_{r-1}) \setminus H, \qquad [x, s_r] \neq 1, \tag{7}$$

modulo conjugation by $C_G(s_1, \ldots, s_r)$. For each orbit representative append x and test the intersection property by the recursive end-parabolic criterion. Keep precisely those extensions that pass. The centralizers in (7) and their orbits are calculated in the explicit forty-point group.

This procedure is exhaustive. A string C-marking is irredundant, so its next generator cannot lie in H. The string relations are exactly the centralizer condition in (7), and conjugating by the pointwise stabilizer of the prefix preserves every extension property. A marking for a nonabelian simple group must be connected: a missing bond would split its generators into nontrivial commuting normal subgroups. Its every prefix is itself a C-marking, so discarding a failed prefix cannot remove a successful full marking. Finally every retained extension at least doubles $|H|$.

No irredundant sequence in a group of order 25920 has length more than fourteen, giving a termination bound independent of the observed early termination.

The complete counts are

length	1	2	3	4	5	6–14
retained prefixes	2	13	121	20	2	0
rejected by intersection	0	0	20	25	0	0

(8)

None of the retained prefixes generates G. Therefore there is no connected string C-marking of G at any rank, and simplicity excludes the disconnected ones.

Attribution and exact remaining scope. The symmetry construction, spinor-norm passage and desired rank spectrum come from Brooksbank, *On the ranks of string C-group representations for symplectic and orthogonal groups*, arXiv:1912.05512v1, 11 December 2019, Theorem 1.1 and Sections 2–4. The endpoint method reuses our acquired characteristic-two endpoint proof for L031; the prime-field translation argument, odd-characteristic generation step and the parameter count above are the present contribution. The rank-contraction theorem is the acquired L037 result. The nonorthogonal perpendicular lines give the separate intersection check required by the central quotient; endpoint injectivity alone is not used to infer it. We retain, rather than reprove, the classical facts listed in the first section and the orders of the classical groups used to identify the finite examples. The finite examples and exceptional exclusion are reproofs of assertions in the supplied source. No rank upper bound for $q \neq 3$ and no solution of an open conjecture is asserted here.